

Roll No: _____

Name: _____

Instructions!

- This quiz is an individual activity, you are strictly NOT allowed to talk with your fellow colleagues, even not allowed to share resources i.e. pen, calculator and/or rough sheets.
- Understanding the questions is, part of the quiz. So, none of your queries will be entertained during quiz time.
- Solve all the questions on the provided space in your question paper.
- This quiz will be collected back after 30 minutes.
- Anyone caught in an act of dishonesty would be awarded an "F" grade in this quiz.

Question 01:

[05 Marks]

Choose the best answer:

1. Which statement is TRUE about Stochastic Gradient Descent?
A) Uses one training example per update
B) Uses full dataset for each update
C) Does not guarantee convergence to the global minima
D) Requires no cost function
2. In multivariate linear regression, the hypothesis in vectorized form is:
A) $h\theta(x) = \theta_0 + \theta_1 x_1$
B) $h\theta(x) = \theta^T x$
C) $h\theta(x) = \theta^T x^T$
D) $h\theta(x) = \theta_0 + \theta^T x$
3. If the learning rate is too large, gradient descent may:
A) Converge very slowly
B) Immediately reach global minimum
C) Diverge or fluctuate
D) Reduce feature importance
4. Which is an advantage of the Normal Equation over Gradient Descent?
A) Faster for extremely large datasets
B) Requires feature scaling
C) No need to choose learning rate
D) Needs many iterations
5. In supervised learning, the training data is:
A) Unlabeled
B) Randomly generated
C) Labeled with correct outputs
D) Noise-free

Question 02:

[05 Marks]

1. What is Feature Engineering? Explain its purpose and discuss atleast three commonly used feature engineering techniques. [5]

Feature engineering is the process of transforming raw data into meaningful, informative, and actionable inputs (features) to enhance model accuracy, reduce complexity, and improve performance, often using domain knowledge. It involves selecting, manipulating, and creating new variables

Purpose: The primary goal is to improve model accuracy and, in many cases, make it easier for algorithms to learn patterns, such as transforming non-linear relationships to linear ones.

Techniques: (any 3)

- 1- Imputation: Filling in missing values (e.g., mean, median, mode).
- 2- Encoding: Converting categorical data (e.g., "Red", "Blue") into numerical values using techniques like One-Hot Encoding.
- 3- Feature Transformation: Applying mathematical operations like logs or square roots to fix skewness.
- 4- Feature Scaling: Scaling up or down the feature to a consistent range. After a scaling operation, the continuous features become similar in terms of range.
- 5- Feature Creation: Generating new features from existing ones, such as extracting "Day of Week" from a timestamp or calculating "Area" from "Length" and "Width".
- 6- Handling Outliers: Removing or capping extreme values that can skew model training. The outliers could be handled as missing values and replaced with suitable imputation.

2. Suppose you're training a model and observe this pertren: [5]

No. of iterations	Cost/loss
100	150
200	170
300	210

Explain what is happening and what is the likely cause of this observation? Also, suggest two solutions to this problem.

The cost is increasing instead of decreasing.

Possible reason: Learning rate is too large. The model is overshooting the minimum and diverging.

Solutions: 1- Reduce learning rate (α) 2- Apply feature scaling

3. You are training a model with two features, where, [4]

- Feature 1: Salary (20,000 – 2,000,000)
- Feature 2: Experience (1 – 10)

What issue may occur? Explain geometrically how this affects gradient descent.

Issue: Different feature scales.

Geometric Explanation:

Without scaling: 1- Cost function contours become elongated ellipses 2-

Gradient descent zig-zags 3- Converges very slowly

With scaling: 1- Contours become circular 2- Faster convergence

Therefore → apply normalization or standardization.

4. Given hypothesis:

$$h\theta(x) = \theta_0 + \theta_1 x_1$$

Where:

$$\theta_0 = 10, \theta_1 = 35$$

For $x_1 = 4$

(a) Compute predicted value.

[2]

hx

$$h\theta(4) = 10 + 35(4)$$

$$= 10 + 140$$

$$= 150$$

(b) If actual value is 20, compute squared error. (mean square error)

[2]

$$(x - y)^2$$

n is total no. of inputs

here $n = 1$ as x_1 is 4 (only 1 input)

$$= 1 / 1 (150 - 20)^2$$

$$= (130)^2$$

$$\text{MSE} = 16900$$

5. You are given the following dataset:

[5]

x	y
1	3
2	5
3	7
4	9
5	11

Using the Normal Equation, compute the optimal values of θ_0 and θ_1 .

solve by using :

$$\theta = (X^T X)^{-1} \cdot X^T y$$

Ans: $\theta_0 = 1, \theta_1 = 2$